Repetition rate function

Description

Function for calculating the repetition rate according to Sproat (2014).

Session Info

Give the session info (reduced).

```
## [1] "R version 4.4.0 (2024-04-24)"
```

```
## [1] "x86_64-pc-linux-gnu"
```

Function

The function takes the parameter "data" as input. This has to be a vector of elements of a sequence. Also, it takes a "method" argument which has to be either "lsmo" (the normalizing R is then the "length of sequence minus one"), or "sproat" (the normalizing R is then the sum of frequency counts *for each type minus 1*).

```
rrate <- function(data, method = c("lsmo", "sproat")){
  # create data frame with frequency counts
  freq.df <- as.data.frame(table(data))
  if (method == "sproat"){
    # get the overall number of repetitions (i.e. adjacent and non-adjacent)
    # as the sum of frequency counts *for each type minus 1*.
    R <- sum(freq.df$Freq-1)
  } else if(method == "lsmo") {
    # get the possible number of repetitions as the length of the sequence minus one.
    R <- length(data)-1
  } else {
    print("Error: wrong method value")
  }
  # calculate the number of adjacent repetitions "r" along the sequence
  r = 0
  if (length(data) > 1){
    for (i in 1:(length(data)-1)){
      if (data[i] == data[i+1]){
        r = r + 1
      } else {
        r = r + 0
      }
    }
  } else {
    r = r
  }
```

```
  # calculate the repetition measure ("r/R" in Sproat)
  # note: since the ratio is not defined for R = 0 (will give NaN),
  # we here hard code rrate = 0 in this case
  if (R == 0){
    rrate = 0
  } else {
    rrate = r/R
  }
  #print(rrate)
}
```

Examples

```
sequence1 <- c("a")
sequence2 <- c("a", "a")
sequence3 <- c("a", "b")
sequence4 <- c("a", "b", "a")
sequence5 <- c("a", "a", "a", "a")
sequence6 <- c("a", "a", "b", "a")
sequence7 <- c(".","*","*",":", ",", ",", ";", ".")

print(rrate(sequence1, method = "lsmo"))
```

```
## [1] 0
```

```
print(rrate(sequence2, method = "lsmo"))
```

```
## [1] 1
```

```
print(rrate(sequence3, method = "lsmo"))
```

```
## [1] 0
```

```
print(rrate(sequence4, method = "lsmo"))
```

```
## [1] 0
```

```
print(rrate(sequence5, method = "lsmo"))
```

```
## [1] 1
```

```
print(rrate(sequence6, method = "lsmo"))
```

```
## [1] 0.3333333
```

```
print(rrate(sequence7, method = "lsmo"))
```

```
## [1] 0.2857143
```

```
print(rrate(sequence1, method = "sproat"))
```

```
## [1] 0
```

```
print(rrate(sequence2, method = "sproat"))
```

```
## [1] 1
```

```r
print(rrate(sequence3, method = "sproat"))
```

```
## [1] 0
```

```r
print(rrate(sequence4, method = "sproat"))
```

```
## [1] 0
```

```r
print(rrate(sequence5, method = "sproat"))
```

```
## [1] 1
```

```r
print(rrate(sequence6, method = "sproat"))
```

```
## [1] 0.5
```

```r
print(rrate(sequence7, method = "sproat"))
```

```
## [1] 0.6666667
```

Literature References

Sproat, R. (2014). A statistical comparison of written language and nonlinguistic symbol systems. Language, 457-481.